Ex:- How many factors does 726 have.

(i) which are all

(ii) which are div by '2' (or) even factors

(iii) Which are odd

(iv) which are Prime

(v) Which are divisible by '11'

(vi) which are divisible by '121'.

(vii) which are divisible by '3'.

(viii) Sum of factors

(ix) No. of ways into product of '2' and factors

(x) Product of factors.

Ans:- $726 = 2 \times 363 \Rightarrow 726 = 2^1 \times 3^1 \times 11^2$

$= 2 \times 3 \times 121.$

$= 2 \times 3 \times 11^2$

(i) All factors $= (1+1)(1+1)(2+1) = 12$

$[2^1, 3^1, 2^1 \times 3^1, 11^1, 11^2, 2^1 \times 11^1, 3^1 \times 11^1, 2^1 \times 3^1 \times 11^1, 2^1 \times 11^2, 3^1 \times 11^2, 1, 2^1 \times 3^1 \times 11^2 = 726]$

(ii) Even factors:- Atleast div by 2^1.

In $2^1 \times 3^1 \times 11^2$

Out of 2 factors $(2^0, 2^1)$ we have to consider 2^1 multiples only to find even factors.

Even factors $= 1 \times (1+1)(2+1) = \boxed{6}$

(iii) Odd factors:- Total factors - Even factors

$= 12 - 6 = \boxed{6}$

(24) (iv) Prime factors:- How many bases of $a^m.b^n.c^0 \ldots$ that many prime factors.

$726 = 2^1 \times 3^1 \times 11^2$

Prime factors = 3 (2, 3 & 11)

(v) factors div by '11':- [at least multiplied by 11^1]

$726 = 2^1 \times 3^1 \times 11^2$

[11^1 & 11^2 to be considered to div by 11]

factors div by '11' :- (1+1)(1+1)(2 (both 11^1 & 11^2 only or $11^0, 11^1$ & 11^2))

$= 2 \times 2 \times 2 =$ 8

(vi) Factors div by '121':- [at least Multiplied by 11^2]

$726 = 2^1 \times 3^1 \times 11^2$ [11^2 only to be considered from $11^0, 11^1$ & 11^2]

factors div by 121 = (1+1)(1+1)(1 (11^2 only)) = 4

(vii) Factors div by '3':- [at least 3^1 Multiplied]

$726 = 2^1 \times 3^1 \times 11^2$ [3^1 only considered of 3^0 & 3^1 for 3]

Factors div by 3 = (1+1)(1 (3^1 only))(2+1) = 6

(viii) Sum of factors:- $\left(\frac{a^{m+1}-1}{a-1}\right)\left(\frac{b^{n+1}-1}{b-1}\right)\left(\frac{c^{0+1}-1}{c-1}\right) \ldots$

Sum of factors = $\left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{3^{1+1}-1}{3-1}\right)\left(\frac{11^{2+1}-1}{11-1}\right) = \frac{3}{1} \cdot \frac{8}{2} \cdot \frac{1330}{10}$

$= 3 \times 4 \times 133 = 532 \times 3 =$ 1596

(ix) No. of ways to Product of 2 and factors:-

Here factors are even then ways = $\frac{\text{factors}}{2} = \frac{12}{2} =$ 6 ways

(x) Product of factors:- Factors are even then

Product = $(726)^{\text{factors}/2} = (726)^6$

↳ Finding Unit digits of given Numbers/Powered No's:-

Unit digit of Base	Cyclic Pattern Number (CPN)	Power / CPN	Remainders	Required Unit digits.
0				0
1				1
2	4	P/4	1, 2, 3 & 0	2, 4, 8, 6
3	4	P/4	1, 2, 3 & 0	3, 9, 7, 1
4	2	P/2	1 [odd power.] 0 [even power]	4, 6
5				5
6				6
7	4	P/4	1, 2, 3, 0	7, 9, 3, 1
8	4	P/4	1, 2, 3, 0	8, 4, 2, 6
9	2	P/2	1 [odd power] 0 [even power]	9, 1

Ex: What is unit digit of $(543)^{87} \times (645)^{55} \times (787)^{85}$

Ans: for '3' $\left(\frac{87}{4} \rightarrow \text{Rem} \rightarrow 3 \Rightarrow 3^3 \rightarrow \text{u.d} \rightarrow 7\right)$.

for '5' (always 5)

for '7' $\left(\frac{P}{4} = \frac{85}{4} \Rightarrow \text{Rem} \rightarrow 1 \Rightarrow \text{u.d} \rightarrow 7\right)$.

$\Rightarrow$ u.d $= 7 \times 5 \times 7 = \underline{5}$

Note:- When 5^x is in finding unit digit of a number and one of the number base in that product is even, then unit digit is always zero.

Examples:- $5^{85} \times 7^{64} \times 3^{81} \times 2^{84} \rightarrow$ '0' u.d.

$6^{64} \times 7^{83} \times 2^{83} \times 8^{65} \times 5^{63} \rightarrow$ u.d $\rightarrow$ '0'.

Ex-4:- $5^{85} \times 7^{43} \times 21^{84} \times 3^{83}$ unit digit is ?

5 ; $\frac{43}{4} \rightarrow R \rightarrow 3 \rightarrow$ '3' ; 1 ; $\frac{83}{4} \rightarrow R \rightarrow 3 \rightarrow$ '7'

u.d $= 5 \times 3 \times 1 \times 7 = 105$

Note:- In the number all bases are odd & one of them is '5', then unit digit of this number is '5' only.

Examples:- $3^{83} \times 1^{84} \times 3^{83} \times 5^{63} \rightarrow$ u.d $\rightarrow 5$

$9^{83} \times 7^{43} \times 3^{185} \times 5^{85} \rightarrow$ u.d $\rightarrow 5$

$1^{85} \times 3^{45} \times 5^{88} \times 7^{84} \times 9^{143} \rightarrow$ u.d $\rightarrow$ '5'.

→ Finding Highest Power:-

Example:-5:- What is highest power of 3 when it devides 500! ? (or)

what is value of x in $\frac{500!}{3^x}$?

↳ In this concept generally if highest power of a prime number is asked, then following procedure to be considered.

3 | 500
3 | 166
3 | 55
3 | 18
3 | 6
2

'x' value (or) highest power is sum of all quotients other than 500 when divided by given prime number [base of number]

$x = 166 + 55 + 18 + 6 + 2 = 249$